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# Increasing the Robustness of Random Bin Picking by Avoiding Grasps of Entangled Workpieces

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## Abstract

In bin picking applications, a robot often picks workpieces that have a complex geometry. This complex geometry can cause entanglements between workpieces resulting in failed grips. This paper presents a machine learning approach to avoid these situations and therefore improves the calculation of suitable grips. Using the depth map of the workpieces and their surrounding neighborhood, a convolutional neural network, which is trained on simulated data, predicts whether an entanglement is present. This information is used to select and calculate the most reliable grip. By avoiding such entangled workpiece situations the robustness of random bin picking increases.

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## 1. Introduction

Bin picking is a common task in most productions. The workpieces are commonly stored chaotically in bins because of the simple solution for storage and transport without the need for specialized workpiece carriers. However, in order to further process the individual workpieces they need to be separated. In the past, this was often done manually or by mechanical feeders like bowl feeders. For the last few years, there has been an immense development of robotic solutions [1]–[4]. For the most common workpiece geometries, the robot systems work quite robust and reliable but there are also workpiece geometries which can entangle. An example for a workpiece geometry with possible entanglements is a connecting rod, whose head can be stuck in an eye of another connecting rod. If a robot picks an object entangled with other objects, the robot's gripper may accidentally pick more than one object at the same time.

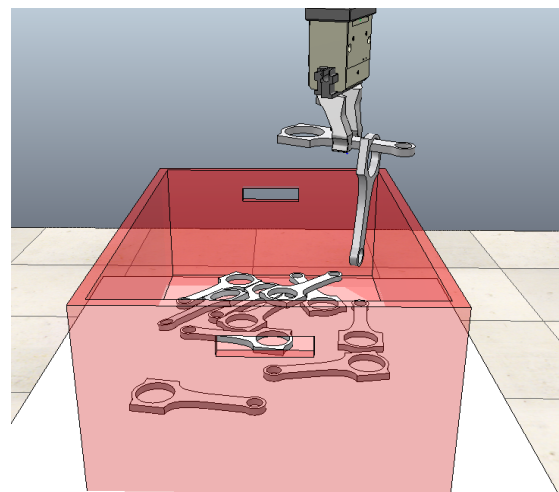


Fig. 1. Simulation scene with a typical entangled workpiece situation.

This would end in a failed grip and therefore in a loss of gripping efficiency and robustness.

Considering a statistic analysis of two random bin picking robot cells, which are separating connecting rods in a production line, there are 591,025 grips logged. 50,718 of these grips failed due to entangled workpiece situations, which can be seen in a simulated scene in Figure 1. This corresponds to an average percentage of 8.6 percent grips of entangled workpieces in different bin fill levels.

To cope with this problem this paper proposes a new learning based approach for detecting entangled workpiece situations. Section 2 discusses current state of the art methods and bring the proposal in context to a bin picking solution in Section 3. Section 4 describes the method concerning the generation of simulated data samples and the training of the convolutional neural network. The results are presented in Section 5 and a conclusion and outlook in Section 6.

In summary, the main contributions of this paper are:

- a method for detecting entangled workpiece situations
- a simulation suite for creating annotated data and a training of a convolutional neural network architecture
- an evaluation of the method on test data samples

## 2. State of the Art

Model-based bin picking consists of two different challenges: the localization of workpieces in a bin and the collision-free gripping with a robot arm. Focusing on the second task, there are different approaches for finding gripping points on workpieces. Some of the methods are limited to specific object geometries [11] or lack flexibility for many industrial applications [12], [13]. Other methods try to avoid using knowledge of the workpiece geometries [14] or focus on learning-based methods [15] - [18]. However, these approaches require extremely large numbers of training data, which is disadvantageous concerning industrial ad-hoc use cases. Furthermore, those proposals are not robust enough for industrial use of picking only one type of object out of a pile. In the proposal of Domae et al. [14] possible grasping poses are determined directly from the depth image of the pile. However, it is impossible to detect an entangled workpiece situation in the pile with this method. Spenrath and Pott [5] set up a best-first heuristic search for robust gripping poses. Depending on the estimated object pose of the workpiece in the pile and various parameters they analyzed in [6], a collision-free gripping pose is calculated. However, the heuristic search does not detect entangled situations.

Concerning the picking of potentially entangled workpieces, there are two publications [9], [10] describing the manipulation of ropes. To the authors best knowledge, there is only one method for detecting entangled workpieces in a pile. Matsumura et al. [7] consider different object geometries in different piled situations and center the gripping points in those. There are depth images rotated so that the parallel jaw gripper of the robot is always oriented horizontally and a convolutional neural network is trained. However, considering only gripping points is not usable for most bin picking systems because of the upstream object pose estimation and the search

for the most reliable object to pick. The main cause for entanglements between objects is the geometry of the object itself. Furthermore, in future developments, there is an opportunity not only to avoid gripping workpieces in those entangled situations but to unhook them, which is only possible by knowing the object geometry. For that reason this paper focusses on the object geometry and the surrounding neighborhood of the workpiece in a pile. To the best of the author's knowledge, this research is the first approach on detecting entangled workpiece situations in a pile focusing on the workpiece geometry itself.

## 3. System Overview

The entanglement detection presented in this paper was developed for model-based bin picking applications using 3D sensor data. Such a bin picking application uses an object localization algorithm to estimate the pose of some of the workpieces within the point cloud or depth map. The score determined by the entanglement detection can then be used to assess the risk of entanglement for localized workpieces and avoid these workpieces in the grasp planning (Figure 2).

A simple way to avoid picking potentially entangled workpieces would be to define a fixed threshold. However, this might block too many parts and does not differentiate between several possible grasps which meet the threshold. Alternatively, the score can be used to prefer grasps on workpieces which have a lower risk of entanglement over other grasps. [5] presented an approach to determine grasps for a model based bin picking application using a heuristic search algorithm. The approach represents all elements used in bin picking (like workpieces, TCPs, grippers, etc.) in a search tree and uses best first search to quickly find a suitable grasp. During this search, collision checks with the point cloud are performed to ensure a collision free grasp. To be able to find a solution quickly, the algorithm uses heuristics to estimate, which elements quickly lead to a solution. For workpieces, this includes e.g. the height of the workpiece inside the bin, since higher workpieces usually are easier to grasp. A localization score, determined by the object localization algorithm can be included as well, to prefer reliably detected workpieces. To be able to avoid picking workpieces, which are potentially entangled, the entanglement score calculated by the approach presented in this paper can be included in the heuristic function as well.

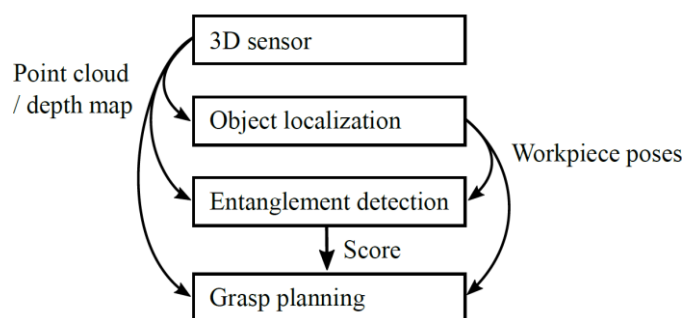


Fig. 2. System context to use the proposed method for bin picking with necessary input data and provided output information.

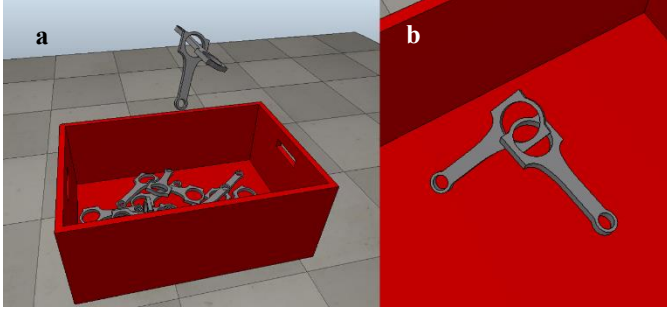


Fig. 3. (a) Provocation of entangled workpiece situations due to dropping already entanglements in the bin; (b) unrealistic entangled situation in the bin.

#### 4. Proposed Method

This section explains the method for detecting an entangled workpiece situation in a pile. Firstly, the annotating metric in a simulation environment to generate our training data set is described. Afterwards, insight is given into the convolutional neural network architecture and the training parameters and datasets.

##### 4.1. Physics Simulation

For detecting entangled workpiece situations in simulated workpiece piles, various physics simulation scenes are set up. Firstly, the focus is on generating bins filled with the chosen object geometry of a connecting rod. Therefore, the dynamic treatment of the physics engine by decomposing the concave shape of the object into several connected convex shapes is improved. To provoke entangled situations in the bin, which can be seen in Figure 3(a), the workpieces are randomly dropped in an already entangled manner. Four categories of bin fill levels are declared, to cover almost every situation in the dataset with different kinds of background noise. There are bins filled with 2, 10, 20 and 40 objects. For further usage of the data, depth images of the filled bin are collected and the homogenous transformation matrix describing the object pose of all the workpieces is logged.

The filled bins are then loaded into another simulation scene for annotating entangled situations. Therefore, the workpieces are set in the same poses as in the bin filling scene and all workpieces start to lift successively of the pile over the bin. Before and right after the lifting, there is a comparison of the object poses of all workpieces  $i$  except the purposely lifted workpiece  $x$ . To validate different displacements in the filled bin while lifting, the Euclidean distances (1) of the object poses are calculated before lifting  $(q_1^i, q_2^i, q_3^i)$  and right after  $(p_1^i, p_2^i, p_3^i)$  of the workpiece  $x$ .

$$|\overrightarrow{PQ}|^i = \sqrt{(q_1^i - p_1^i)^2 + (q_2^i - p_2^i)^2 + (q_3^i - p_3^i)^2} \quad (1)$$

For the lifted object all Euclidean distances are added to a total  $S^x$  (2) and the totals are normalized after lifting all workpieces in the bin to an unrest label  $L^x$  (3) with the maximum  $S^{max}$  and minimal total  $S^{min}$  of the respective bin scene.

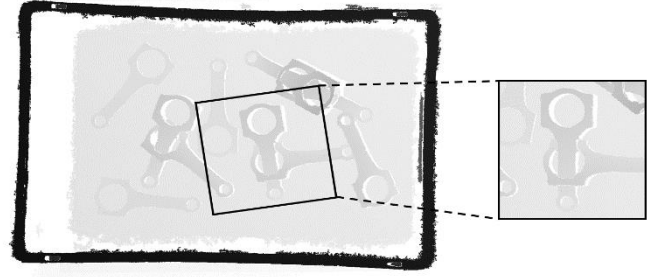


Fig. 4. Cropping the depth image of the whole scene to a depth image orientated along a coordinate axis of the workpiece.

$$S^x = \sum_{i \in \{x\}} |\overrightarrow{PQ}|^i \quad (2)$$

$$L^x = 1 - \left[ \frac{(S^x - S^{min})}{(S^{max} - S^{min})} \right] \quad (3)$$

However, there are some unrealistically entangled workpiece situations as shown in Figure 3(b) due to failures in the dynamic consideration regarding convex decomposition shapes of the physics engine. To eliminate those unrealistic situations, the lifted workpiece is rotated around all coordinate axes. If the entanglement is still present after rotating, the data sample is discarded as an error. Each workpiece will then be binary annotated and classified through fixed thresholds around the lifting height of maximum normalized labels  $L^x$ . After each lift, the situation in the bin is restored to the original situation.

Most object pose estimation metrics use the depth image or point cloud of the whole bin scene and do not find all present workpieces in the scene. Therefore, the depth image of the whole scene is cropped into several depth images for every workpiece in the same manner. An example is shown in Figure 4. To achieve this, the metric use the object pose matrix given from the bin filling scene, project the object into the x-y-plane and orient the crop along a specific coordinate axis. By this depth image preparation, every object pose estimation metric which finds only a few of the present workpieces can use this approach. Every depth image crop is then paired with the corresponding binary label of predicted entanglement.

##### 4.2. Entanglement Detection Method

For prediction of possible entangled workpiece situations in a pile, a convolutional neural network was trained on simulated data. The input given to the network is a cropped depth image with 160x160 pixels from the top view of the bin oriented along a certain coordinate axis of the selected workpiece. Thus, the focus of the network training will be on the immediate neighborhood of the workpiece. The architecture of the convolutional neural network is inspired by the DenseNet [8] network, which strengthens feature propagation and encourages feature reuse and is shown in Figure 5.

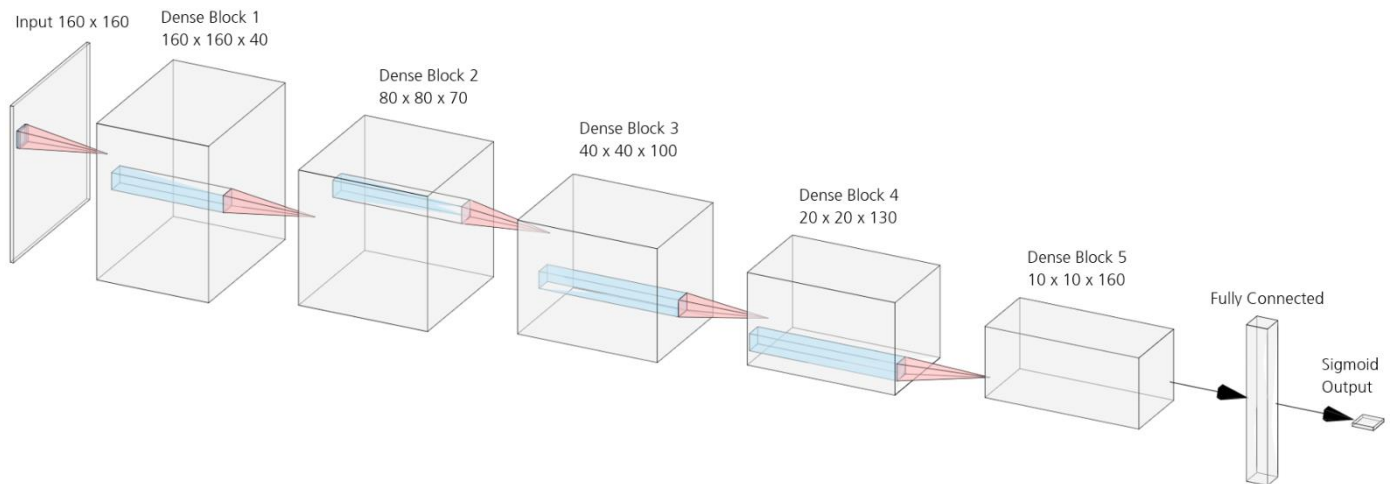


Fig. 5. Architecture of the densely convolutional neural network used for the classification of cropped depth images concerning if an entangled workpiece situation is present or not

The convolutional neural network is composed of five dense blocks, each consisting of multiple convolutional layers. The first dense block starts with four convolutional operations while following dense blocks include only three. Every convolutional layer inside of a dense block gets the output information from every previous convolutional layer as input. Between these blocks, an average pooling with 2x2 pixels is applied. One fully connected layer, which classifies whether there is an entangled workpiece situation or not, processes the result. As activation function of the output layer, a sigmoid-function is used, whereas every other layer uses a rectified linear unit function. The network was trained for 100 epochs

on a training data set of around 17,500 data samples, gathered from our simulation with a batchsize of 256. The validation set contains nearly 4,000 unseen data samples, which are divided into five different categories. One training epoch takes an average training time of about 30 seconds on a NVIDIA Tesla V100 PCIe 32GB GPU. Figure 7 shows the decreasing loss and the increasing accuracy during training of the neural network. The training data set is completely shuffled with entangled and non-entangled workpiece situations with different counts of workpieces in the pile to gain different variants of background noise because of other workpieces in the scene.

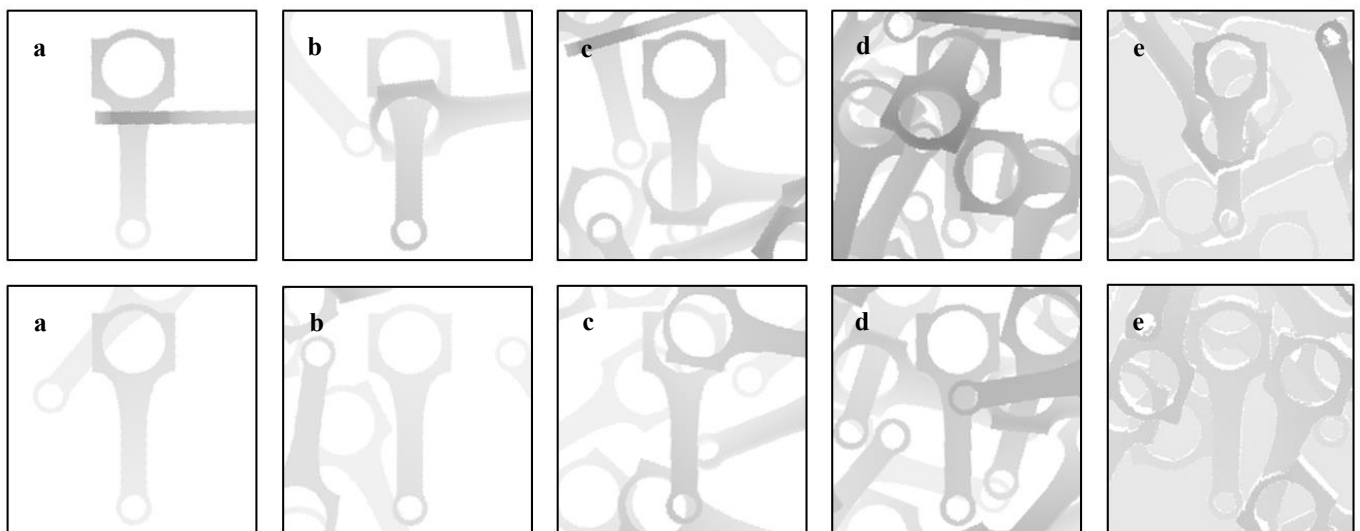


Fig. 6. Test set with (a) cropped depth images of entangled workpiece situations (top) and non-entangled workpiece situations (bottom) in category 2 objects; (b) ...category 10 objects; (c) ...category 20 objects; (d) ...category 40 objects; (e) ...category variable count of objects in real world



## 5. Results

Table 1 shows the results of the trained neural network in different test scenes. The test data set is divided into five different categories with entangled and non-entangled data samples as shown in Figure 6. In the first category, there are only piles of two workpieces in the scene to get data samples with less neighborhood noise. All entanglements and all non-entangled situations were detected by the neural network with an average accuracy of 100 percent. The same results are provided in category two, where ten workpieces are present in a filled bin. Here, there is more neighborhood and background noise than in the first category, but the network is predicting all situations correctly. However, the neural network starts to get less accurate the more workpieces are in the scene. In the third category, there are 20 workpieces dropped in the bin, so that the bottom of the bin is almost covered. For this fill level, the neural network predicts the entangled workpiece situations with an average accuracy of 95 percent. To bring it into counts, the network detected 911 situations of a total of 1,054 situations correctly. There are 32 false positive detections which means, that the neural network proposes an entangled situation but there is none. These predictions often happen in difficult situations where many workpieces are on top of the concerned workpiece. In this case, the probability to grasp one of these candidates is very low because of the restrictions of several gripping point searches or heuristics. However, in these uncertain situations it is better to propose a false positive and avoid gripping those workpieces. In category four, there are 40 workpieces dropped into the bin and evaluated the entanglement prediction. The neural network achieved an average accuracy of 86 percent and an average precision of 71 percent. Only 11 situations out of 950 samples are detected as false negatives, which means the robot would pick an entangled workpiece. To compare the given numbers of false negatives to the statistics in the first section of the paper, the method would achieve more than 8 times less failed grips due to entanglements than the state of the art, if the filled bin level is about 40 workpieces and there is a lot of noise in the background because of other workpieces.

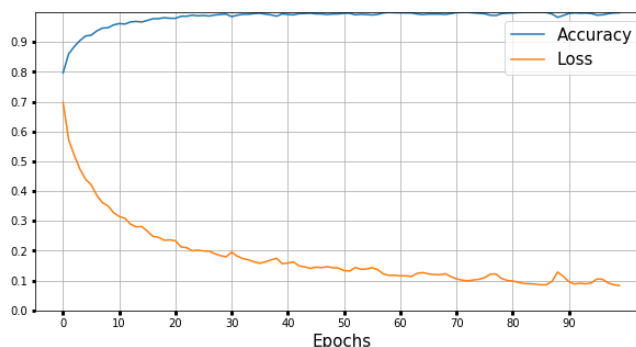


Fig. 7. Diagram of increasing accuracy and decreasing loss during training of the densely convolutional neural network.

In the last category, the focus is on manually labeled real world data, recorded with an Ensensio X36 stereo vision camera. The optical system works in normal lighting conditions through to a projection module in the middle of the device. The system is mounted in a distance of 1,500mm to the recorded objects in the scene. The cropped depth images are classified with our neural network completely trained on simulated data. The achievement is an average accuracy of 88 percent on unseen situations and an average precision of 95 percent. These correlations prove that the trained neural network is robust enough to be implemented in real robot cells to avoid entangled workpiece situations. Due to training on real world data samples, the accuracy and precision would be increased.

All predictions were calculated on a NVIDIA Tesla V100 PCIe 32GB GPU within a mean average prediction time of about 8.5 milliseconds. The mean average accuracy amounts to 91.2 percent with respect to all testing categories. The mean average precision is also about 91.2 percent concerning all kinds of bin fill levels for simulated and real data samples. 1,313 of 1,345 entangled workpiece situations in the test data set were detected correctly. Out of 3,793 workpiece situations, only 32 would result in a failed grip, which decreases the failure rate from 8.6 percent due to entanglements to about 0.8 percent.

Table 1. Validation results on the test data set divided into 5 different categories (in brackets the total count of specific situations).

| Category                                | True positives | True negatives | False positives | False negatives | Average accuracy | Average precision | Average recall |
|-----------------------------------------|----------------|----------------|-----------------|-----------------|------------------|-------------------|----------------|
| simulated data, 2 workpieces each pile  | 354 (354)      | 550 (550)      | 0               | 0               | 1.0              | 1.0               | 1.0            |
| simulated data, 10 workpieces each pile | 328 (328)      | 558 (558)      | 0               | 0               | 1.0              | 1.0               | 1.0            |
| simulated data, 20 workpieces each pile | 295 (306)      | 616 (648)      | 32              | 11              | 0.95             | 0.90              | 0.96           |
| simulated data, 40 workpieces each pile | 295 (306)      | 526 (644)      | 118             | 11              | 0.86             | 0.71              | 0.96           |
| real data shuffled, different counts    | 41 (51)        | 46 (48)        | 2               | 10              | 0.88             | 0.95              | 0.8            |
| Mean averages over all categories       |                |                |                 |                 | 0.94             | 0.91              | 0.94           |

## 6. Conclusion and Outlook

In this paper, there is a method for detecting an entangled workpiece situation in a pile of workpieces in random bin picking proposed. A convolutional neural network is trained on an own dataset of entangled workpiece situations, which was annotated with several implemented metrics. As a conclusion, there is confirmation that using this method after an object pose estimation will detect entanglements in a pile and the proposed entanglement score can be used to avoid picking workpieces out of these situations. In summary, this method helps to increase the robustness and efficiency of random bin picking by avoiding grasps of entangled workpieces with the usage of the proposed entanglement score.

In future work, the method will be implemented to a real robot cell with a vision based validation to distinguish between entangled workpieces in the gripper and valid grips. This information will be used for further fitting of the neural network in an online training framework. Furthermore, based on concentrating on the cropped workpiece depth image, there will be work on a method to not only avoid those entangled situations, but to also solve and unhook them. Therefore, there is need of the detection and the data basis presented in this work.

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